Brain MRI Segmentation under Class Imbalance: A Comparative Study of Machine and Deep Learning models

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***Abstract*—This study investigates both classical machine learning approaches and modern deep learning pipelines based on U-Net architectures for the segmentation of anatomical regions in brain images. The deep learning models achieve segmentation dice scores exceeding 80 across all classes, whereas classical approaches struggle to surpass 50 accuracy for rare structures such as the amygdala and hippocampus. In particular, Random Forest classifiers are evaluated under multiple configurations, including hyperparameter search and cross-validation across key preprocessing steps, to assess their impact on segmentation performance. A major challenge is highly imbalanced class distributions, with rare anatomical regions being especially difficult to segment accurately. To address this issue, class weighting strategies are applied for machine learning based models. Segmentation performance is primarily evaluated using the Dice coefficient, which is observed to deteriorate as higher weights are assigned to rare classes due to an increase in false positive predictions. In contrast, recall increases under class-weighted training, as these configurations reduce false negative errors and improve sensitivity to rare anatomical structures. U-Net–based deep learning models, however, whether trained from scratch or fine-tuned from foundation models, demonstrate consistently strong performance and show limited sensitivity to class imbalance and low-data regimes, highlighting their superior capacity to learn robust and discriminative representations for both common and rare anatomical structures.**

***Index Terms*—Brain Image Segmentation, Random Forest, U-Net, Class Imbalance, Foundation Models, Dice Coefficient, Recall**

I. Introduction

The rapidly growing volume of medical imaging data has created a strong demand for automated processing methods to support downstream clinical tasks such as diagnosis, treatment planning, intraoperative monitoring, and postoperative assessment. Brain medical imaging is no exception to this trend. In particular, accurate segmentation of anatomical brain regions remains a fundamental and challenging problem, as it constitutes a prerequisite for many subsequent analysis tasks by delineating distinct anatomical boundaries.

To address this objective, voxel-wise classification approaches for brain image segmentation have received considerable attention. Early work in this domain primarily relied on classical machine learning (ML) methods, including k-nearest neighbors [1], support vector machines [2], and Random Forest classifiers [3], often combined with handcrafted features derived from image intensity, spatial context, and atlas-based information [4] [5]. While these methods have demonstrated reasonable performance, they typically require careful feature engineering and are sensitive to class imbalance, particularly when segmenting small anatomical structures.

More recently, deep learning methods have become the dominant paradigm in medical image segmentation. Convolutional neural networks, and in particular the U-Net [6] architecture, have shown remarkable success by learning hierarchical and spatially rich feature representations directly from the data. Subsequent extensions and variants of U-Net have further improved segmentation accuracy and robustness, reducing the reliance on handcrafted features and enabling end-to-end training for complex segmentation tasks.

In this work, we explore both classical and modern approaches for voxel-wise brain region segmentation.

Through this comparative study, we aim to analyze the strengths and limitations of classical machine learning and modern deep learning approaches for brain image segmentation, with particular emphasis on performance under class imbalance and the segmentation of small anatomical structures. Furthermore, within the Random Forest–based pipeline, we investigate the robustness of different evaluation metrics to the use of class weighting. In particular, we compare the Dice coefficient and recall to assess how segmentation performance changes after applying class weighting strategies, and to better capture improvements that may not be reflected by Dice score alone. For deep-learning models we investigate the training models from scratch and also training from pretrained models which are called Foundation models.

II. Methodology

A. Dataset

The experiments are conducted on a dataset consisting of T1- and T2-weighted brain MRI scans with corresponding voxel-wise ground truth labels. A total of 20 scans are used for training, while 7 additional scans are reserved for testing.

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The labeled regions include both large and small anatomical structures, resulting in a highly imbalanced class distribution, particularly for rare regions such as the amygdala, Thalamus and hippocampus as summarized in Table I.

TABLE I
REGION STATISTICS WITH IMBALANCE ASSESSMENT

Region	Pixel Count	Imbalance Score
background	45817	0.483
WhiteMatter	38969	0.568
GreyMatter	31234	0.709
Hippocampus	7091	3.123
Amygdala	2598	8.525
Thalamus	7171	3.088

B. Preprocessing

All images undergo a standard preprocessing pipeline prior to segmentation. Each scan is registered to a reference atlas to ensure spatial alignment across subjects. Intensity values are normalized using z-score normalization to reduce inter-subject variability. In addition, a brain mask is applied to remove background voxels and restrict the analysis to brain tissue only. These preprocessing steps are applied consistently across all classical machine learning methods, however, deep learning pipelines apply only z-scoring on the pixel level.

C. Random Forest Implementation

1) Feature Extraction: For the ML–based segmentation pipeline, voxel-wise feature vectors are constructed by combining spatial and local intensity-based features. Atlas-based spatial features consist of the physical atlas coordinates (x, y, z), which encode anatomical location after registration. Local first-order texture features are computed within a three-dimensional neighborhood and include mean, variance, standard deviation, skewness, kurtosis, entropy, energy, signal-to-noise ratio (SNR), minimum, maximum, range, and intensity percentiles (10th, 25th, 50th, 75th, and 90th). The final feature vector is obtained by concatenating spatial and texture features for each voxel.

2) Random forest Configurations: Several Random Forest configurations are evaluated to assess the impact of model complexity and training strategies on segmentation performance. Starting from a baseline model with default parameters. We progressively incorporate hyperparameter search with cross-validation, class weighting, and feature normalization. The evaluated configurations differ mainly in the number of trees and maximum tree depth, allowing us to analyze the effect of increased model capacity and imbalance-aware training on segmentation performance. Table II summarizes the Random Forest variants considered in this study.

The default configuration serves as a baseline with minimal model complexity. The random search configuration explores a broader hyperparameter space to identify more suitable values for tree depth and ensemble size. The random search with cross-validation further improves robustness by selecting hyperparameters that generalize better across data splits. To address the strong class imbalance present in brain region segmentation, class weighting is introduced to emphasize rare anatomical structures during training. Finally, feature normalization is combined with class weighting to reduce scale differences between features and to assess its effect on classifier stability and performance.

To further tackle the imbalanced dataset, Balanced Random Forest from Imbalance Learn Library [7] which is part of Scikit Learning main project is used. Unlike standard random forest, balanced random forest adjusts the sampling process by taking a bootstrap sample from the minority class and then drawing a matching number of samples with replacement from the majority class. The rest of the model algorithms are the same as the conventional random forest.

TABLE II
RANDOM FOREST CONFIGURATIONS

Approach	Max depth	n-estimators
Default Values	1	5
Random Search	19	89
Random Search + Cross Validation	57	17
Random Search + Class Weight	57	17
Random Search + Class Weight + Feature Normalization	57	19

3) Extension to Extremely Random Forests: In addition to the mentioned Random Forest classifiers, we explore an extension based on Extremely Randomized Trees (ExtraTrees) [8] to investigate whether increased randomness in tree construction can improve segmentation performance, particularly for imbalanced class distributions. Unlike conventional Random Forests, where split thresholds are selected by optimizing a local impurity criterion, ExtraTrees select split thresholds at random, which reduces tree correlation and model bias. This additional randomization is expected to improve generalization and reduce overfitting to dominant classes, a common issue in voxel-wise brain segmentation. While ExtraTrees does not explicitly address class imbalance, its increased robustness to noise and reduced bias can lead to improved recognition of rare anatomical structures when combined with class weighting strategies.

D. Implementation of U-Net

The deep learning–based segmentation pipeline employs a U-Net architecture for voxel-wise brain MRI segmentation. The network takes T1- and T2-weighted images as input and produces a multi-class segmentation map as output. U-Net follows an encoder–decoder design with skip connections that combines high-level semantic information with fine-grained spatial details. This architectural design enables precise localization while preserving contextual information, which is particularly important for accurately segmenting small and rare anatomical structures. By learning hierarchical and spatially consistent feature representations directly from the data, the model reduces reliance on handcrafted features and improves generalization across classes with varying sizes. The overall

TABLE III
SUMMARY OF CLASS WEIGHTING CONFIGURATIONS USED IN THE RANDOM FOREST EXPERIMENTS

Config	Weighting Strategy
erf / erf-cv	No class weighting(UW)
erf-cv-1-2-1	Mild emphasis on rare class(MW)
erf-cv-balanced	Inverse-frequency balancing(BRF)
erf-cv-rnd	Randomized class weights(RW)
erf-cv-1	Uniform class weights(UW)

network architecture is illustrated in Fig. 1. The UNet framework for training is nnUZoo library [9] which is based on nnUNet [10] framework. 2D and 3D UNet models are both investigated.

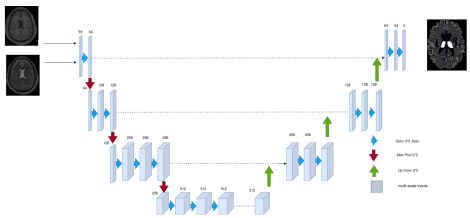

Fig. 1. Overview of the U-Net architecture used for brain MRI segmentation. The network follows an encoder–decoder structure with skip connections that combine high-level contextual features from the contracting path with fine-grained spatial information from the expanding path.

Extension to Foundation Models: In addition to training U-Net models from scratch, we explore the use of foundation models through fine-tuning pre-trained architectures from the OpenMind [11] dataset which are trained using nn-UNet [10] framework. These models are pre-trained on large-scale brain MRI datasets and provide strong initialization for downstream segmentation tasks. The network is fine-tuned using three-dimensional T1- and T2-weighted MRI inputs, producing voxel-wise segmentation outputs. By leveraging representations learned from large and diverse datasets, the foundation model is expected to improve generalization and segmentation performance, particularly in cases with limited training data and a small number of voxels for rare anatomical structures. During the training process of the foundation model, both the contracting and expanding paths of the network are fine-tuned.

E. Evaluation Metrics

The segmentation models were trained using two different Evaluation Metrics to handle challenges such as class imbalance and to refine the focus on difficult regions. The Metrics used are as follows:

Dice coefficient: This is a commonly used metric for medical image segmentation, particularly effective for handling class imbalance by focusing on maximizing the overlap between the predicted and ground truth regions. The Dice coefficient is defined as:

$$\text{Dice}_c = \frac{2TP_c}{2TP_c + FP_c + FN_c} \tag{1}$$

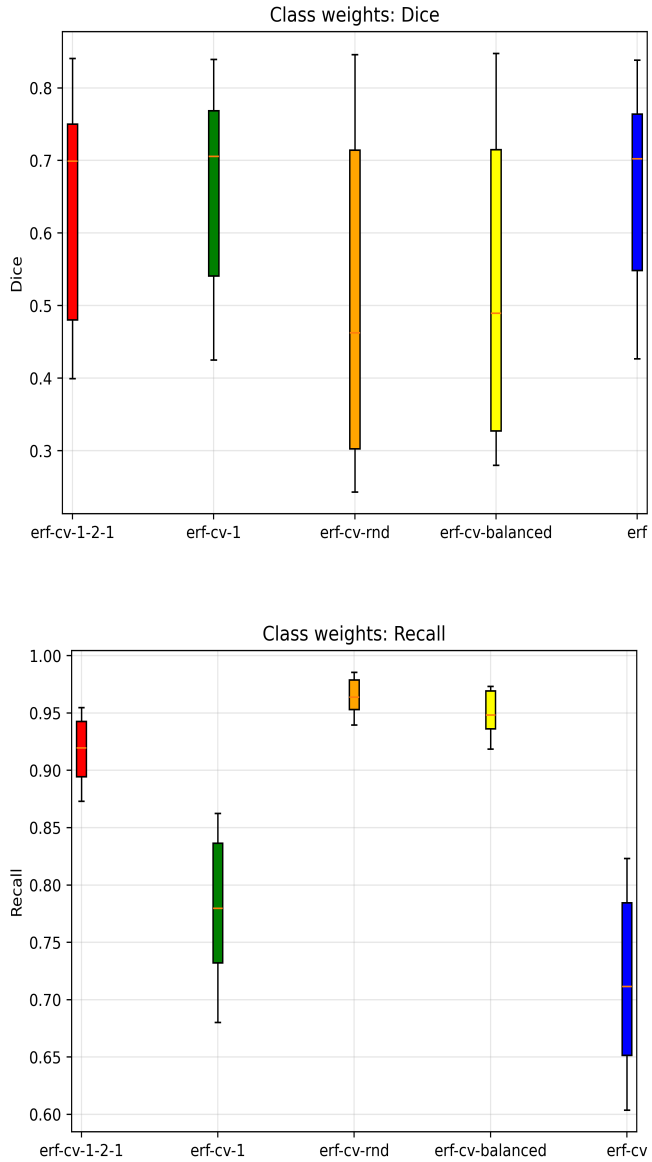

Fig. 2. Performance distributions under different class weighting configurations. Top: Dice coefficient. Bottom: Recall. Configuration mappings are given in Table III.

where TP stands for True Positive, FP for False Positive, And FN for Flase Negetive class predictions. Index c stands for class as we are computing Dice coefficient per class.

Recall: Due to the limitations observed when using the Dice coefficient to reflect performance improvements for rare classes, we additionally consider recall as an evaluation metric. Recall measures the ability of the model to correctly identify voxels belonging to a given class and is particularly sensitive to false negative predictions, which are common for small and underrepresented anatomical structures. Recall is defined as:

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{2}$$

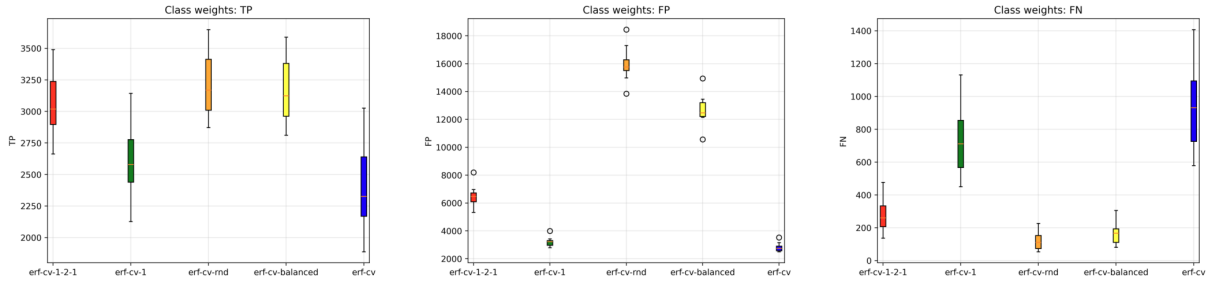

Fig. 3. Distribution of true positive (TP), false positive (FP), and false negative (FN) predictions, shown in the left, middle, and right panels, respectively.

where TPc denotes the number of true positive predictions for class c, and FNc denotes the number of false negative predictions. In the Results section, we show that class weighting strategies tend to reduce false negatives while simultaneously increasing false positives. As a consequence, recall may increase even when the Dice coefficient decreases. This highlights the importance of considering recall alongside Dice when evaluating segmentation performance for rare and small anatomical structures.

III. Results

A. (Extreme) Random Forest

The results for all Random Forest configurations and their Extremely Randomized variants are summarized in Table IV.

B. Class Weighting Configuration

As discussed previously, rare classes corresponding to small anatomical structures pose a significant challenge for voxel-wise classification. To address this issue, we explore several class weighting configurations in order to analyze their impact on segmentation performance. Table V summarizes the class weighting schemes evaluated in this study, while Fig. 2, top, illustrates the distribution of Dice coefficients obtained under each configuration.

From Fig. 2, top, it can be observed that assigning higher weights to rare classes generally leads to a decrease in the Dice coefficient. This behavior can be directly explained by the Dice formulation in Eq. (1) and the effect of class weighting during training. Increasing the weight of a rare class encourages the classifier to predict more voxels as belonging to that class, which often results in an increase in false positive predictions illustrated in Fig 3 (middle). Since false positives appear in the denominator of the Dice coefficient, this increase leads to a reduction in the Dice score, even in cases where the number of true positives increases Fig 3 (left). This highlights a limitation of the Dice metric in reflecting performance improvements for rare classes under class-weighted training.

In contrast, recall tends to increase when class weighting is applied, as shown in Fig. 2, Bottom. This observation is in alignment with Fig 3 (right) showing a reduction in false negative predictions under class-weighted configurations. As false negatives directly appear in the denominator of the recall formulation, this reduction leads to the observed increase in recall according to Eq. (2).

C. Deep Learning (U-Net)

In contrast to Random Forests, which exhibited notable sensitivity to class imbalance, the deep learning–based U-Net model demonstrates substantially more robust segmentation performance across all anatomical classes. By learning hierarchical and spatially contextual feature representations directly from the data, the U-Net is able to accurately classify voxels belonging to both large and small anatomical structures, achieving Dice scores above 0.80 even for rare classes such as the amygdala and hippocampus.

Table VI summarized model and training information of all U-Net models and Table VII summarizes the segmentation performance of the 2D U-Net model and the 3D U-Net model trained from scratch and the fine-tuned foundation model. The results indicate that all training strategies achieve comparable performance across all evaluated brain regions, suggesting that the U-Net architecture itself is highly effective at handling class imbalance in this setting. The use of a foundation model does not yield a significant improvement in Dice scores for this dataset, likely due to the relatively limited complexity of the segmentation task and the strong baseline performance of the scratch-trained model.

Table VIII compares the segmentation performance of all models evaluated in this study. Classical machine learning approaches, including Random Forest and Extreme Random Forest, achieve moderate Dice scores, particularly struggling with small and rare anatomical structures such as the amygdala and hippocampus. In contrast, deep learning–based U-Net models substantially outperform classical methods across all regions. A visual comparison of all the configurations for Random Forest and U-Net models are illustrated in Fig. 4.

Discussion

This work analyzed the impact of class imbalance on brain MRI segmentation by comparing classical Random Forest–based methods with deep learning–based U-Net models, with particular emphasis on the role of class weighting and metric selection. The results show that class weighting substantially affects the behavior of Random Forest classifiers,

TABLE IV

DICE SCORES (MEAN ± STD) FOR DIFFERENT RANDOM FOREST–BASED SEGMENTATION VARIANTS. CONFIGURATIONS ARE DENOTED USING THE FOLLOWING ABBREVIATIONS: EXTREMELY RANDOMIZED FOREST (ERF), FEATURE NORMALIZATION (FN), HYPERPARAMETER-OPTIMIZATION WITH CROSS-VALIDATION (HOCV), AND WEIGHT BALANCE (WBLC)

Label	ERF + FN	ERF + HOCV + FN + WBlc.	ERF + HOCV + FN	RF + HOCV + FN	ERF + HOCV + FN
Amygdala	$\mathbf{0.56 \pm 0.04}$	0.33 ± 0.02	0.55 ± 0.05	0.54 ± 0.04	0.55 ± 0.05
Grey Matter	0.70 ± 0.008	0.70 ± 0.008	0.71 ± 0.008	$\mathbf{0.72 \pm 0.008}$	0.71 ± 0.008
Hippocampus	0.47 ± 0.02	0.32 ± 0.01	0.48 ± 0.02	$\mathbf{0.50 \pm 0.01}$	0.48 ± 0.02
Thalamus	$\mathbf{0.75 \pm 0.02}$	0.49 ± 0.026	0.72 ± 0.03	0.74 ± 0.02	0.72 ± 0.031
White Matter	0.80 ± 0.02	0.80 ± 0.01	0.81 ± 0.01	$\mathbf{0.82 \pm 0.01}$	0.80 ± 0.01

TABLE V

DICE SCORES (MEAN ± STD) UNDER DIFFERENT CLASS WEIGHTING STRATEGIES. RW: RANDOMIZED WEIGHTS (0.5, 0.56, 0.7, 4, 15, 4); UW: UNIFORM WEIGHTS (1,1,1,1,1); MW: MILD WEIGHTING (1,1,1,2,1); BRF: BALANCED RANDOM FOREST WITH CLASS-BALANCED SAMPLING STRATEGY.

Label	Balanced	RW	UW	MW	BRF
Amygdala	0.33 ± 0.02	0.28 ± 0.04	$\mathbf{0.56 \pm 0.03}$	0.47 ± 0.03	0.38 ± 0.03
Grey Matter	0.70 ± 0.00	$\mathbf{0.72 \pm 0.00}$	0.70 ± 0.00	0.70 ± 0.00	0.69 ± 0.00
Hippocampus	0.32 ± 0.01	0.30 ± 0.01	0.46 ± 0.01	$\mathbf{0.47 \pm 0.02}$	0.35 ± 0.01
Thalamus	0.49 ± 0.02	0.46 ± 0.02	$\mathbf{0.74 \pm 0.03}$	0.72 ± 0.03	0.64 ± 0.02
White Matter	0.81 ± 0.01	0.81 ± 0.01	0.81 ± 0.01	0.81 ± 0.01	0.81 ± 0.01

TABLE VI

U-NET MODEL TRAINING CONFIGURATIONS. FM IS FOUNDATION MODEL (FINE-TUNED). ALL MODELS TRAINED WITH SGD OPTIMIZER. TRAINING TIME MEASURED IN MINUTES.

Model	Epochs	GPU Type	lr	Time (M)
2D U-Net	100	RTX 4090	1e-2	100
3D U-Net (scratch)	100	RTX 4090	1e-2	165
3D U-Net (FM)	100	RTX 4090	1e-3	165

TABLE VII

DICE SCORES FOR 3D U-NET SEGMENTATION TRAINED FROM SCRATCH AND USING A FOUNDATION MODEL.

Model	White Matter	Grey Matter	Hippocampus	Amygdala	Thalamus
2D	0.90	0.81	0.85	0.87	0.87
3D Foundation Model	0.91	0.82	0.87	0.84	0.91
3D Scratch	0.91	0.82	0.87	0.84	0.91

TABLE VIII

COMPARISON OF DICE SCORES (MEAN ± STD) ACROSS MODELS. RF: RANDOM FOREST; ERF: EXTREME RANDOM FOREST; U2D: 2D U-NET; U3D: 3D U-NET; FM: U-NET 3D FOUNDATION MODEL.

Label	RF	ERF	U2D	U3D	FM
Amygdala	0.54 ± 0.04	0.55 ± 0.05	0.82	0.84	0.84
Grey Matter	0.72 ± 0.008	0.71 ± 0.008	0.81	0.82	0.82
Hippocampus	0.50 ± 0.01	0.48 ± 0.02	0.85	0.87	0.87
Thalamus	0.74 ± 0.02	0.72 ± 0.031	0.90	0.91	0.91
White Matter	0.82 ± 0.01	0.80 ± 0.01	0.90	0.91	0.91

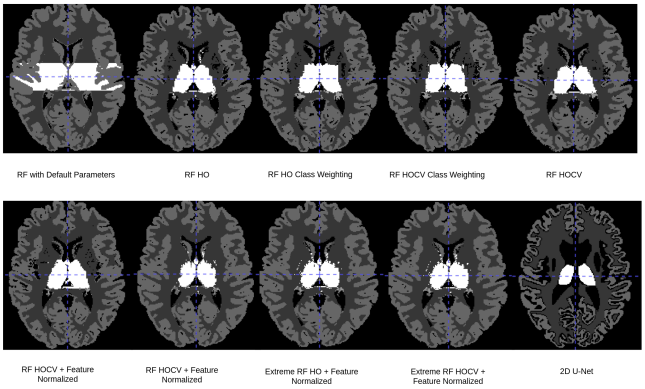

Fig. 4. Visual comparison of segmentation results on a single axial brain slice across different models. While Random Forest–based methods exhibit variability and over-segmentation in deep regions such as the thalamus (central white region), the deep learning–based U-Net model produces more accurate and spatially consistent segmentations across all classes.

especially for small and underrepresented anatomical structures such as the amygdala and hippocampus.

As demonstrated by the Dice and Recall distributions in Fig. 2, top, and Fig. 2, bottom, and the quantitative results in Table V, increasing the weight assigned to rare classes generally leads to a decrease in Dice scores while simultaneously improving recall. This divergence can be explained by the Dice formulation, where an increase in false positive predictions—induced by class-weighted training—negatively impacts the Dice coefficient despite a reduction in false negatives. In contrast, recall directly benefits from reduced false negatives, making it more sensitive to improvements in rare-class detection. These findings highlight an important limitation of relying solely on Dice when evaluating segmentation performance under class imbalance.

The distributional analysis further reveals that moderate or balanced weighting strategies yield more stable performance, whereas aggressive or randomized weighting introduces higher variability. This suggests that while class weighting can enhance sensitivity, it must be applied cautiously to avoid instability and degradation in overlap-based metrics.

In contrast to classical methods, the U-Net–based models

demonstrate strong robustness to class imbalance, achieving consistently high Dice scores across all anatomical regions without the need for explicit class weighting. This robustness is attributed to the ability of deep learning models to learn hierarchical and spatially contextual representations directly from the data. The foundation model, on the other hand, in terms of final segmentation performance remains comparable to the scratch-trained U-Net for this dataset.

From a practical standpoint, the choice between deep learning and classical methods involves significant trade-offs. While deep learning models like U-Net offer superior robustness to class imbalance and state-of-the-art performance, they demand substantial computational resources, including dedicated GPUs for feasible training times, and require larger datasets for effective training. Classical machine learning methods like Random Forest, though potentially less powerful for complex segmentation, offer advantages in computational efficiency, shorter training times on standard CPUs, and greater interpretability, making them viable options in resource-constrained environments or for initial exploratory analysis.

Overall, the results emphasize that class weighting is a double-edged strategy for classical machine learning–based segmentation and that complementary metrics such as recall are essential for a meaningful evaluation of rare anatomical structures. Deep learning–based approaches, by contrast, inherently mitigate many imbalance-related challenges and provide superior and more stable segmentation performance.

Conclusion and future Work

This study demonstrates that deep learning–based segmentation models consistently outperform classical machine learning approaches in brain MRI segmentation, particularly for small and underrepresented anatomical structures. While Random Forest–based methods benefit from techniques such as feature normalization and class weighting, their performance remains sensitive to class imbalance and metric selection. In particular, class weighting does not improve the overall Dice score and should therefore be applied selectively, depending on task-specific priorities. The complementary use of evaluation metrics such as recall provides additional insight into segmentation performance that may not be captured by Dice alone.

Future work will focus on improving both classical and deep learning pipelines. For machine learning approaches, this includes evaluating alternative atlas registration strategies, exploring balanced voxel sampling schemes, and incorporating post-processing techniques. For deep learning models, future directions include investigating transformer-based architectures, extending segmentation to 2.5D models, exploring additional foundation model training strategies, and studying the effect of class weighting within deep learning frameworks.

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